

MERN STACK INTERNSHIP – DAY 6 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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Submission Date: 06-06-2026

FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

Objective

The objective of Day 6 was to review and validate the MERN stack application developed during previous assignments. The focus was on production build verification, routing configuration analysis, database connectivity review, Express middleware inspection, and understanding of frontend state management and data synchronization mechanisms.

Task 6.1 – Code Defense & Refactoring

Production Build Validation

The React application was validated for production readiness by generating an optimized production build.

Build Command

```
npm run build
```

Outcome

- Production build generated successfully.
- Application assets were optimized and minified.
- Distribution folder was created successfully.
- No build-time errors were encountered.

Benefits of Production Build

- Reduced bundle size
- Faster loading performance
- Optimized asset delivery
- Improved deployment readiness

Routing Configuration Review

The application's routing structure was reviewed to ensure proper navigation between pages and components.

Routing Features

- Route-based navigation implemented.
- Client-side routing handled through React Router.
- Component rendering based on URL paths.
- Improved user experience without page reloads.

Example Structure

```
<BrowserRouter>
  <Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
  </Routes>
</BrowserRouter>
```

Database Connection Validation

Backend database connectivity was reviewed using MongoDB and Mongoose.

Features Verified

- Successful database connection.
- Proper use of environment variables.
- Stable communication between backend and database.
- Efficient handling of database operations.

Example Connection

```
mongoose.connect(process.env.MONGO_URI);
```

Express Middleware and Security Review

Middleware configuration was inspected to ensure proper request handling and backend stability.

Middleware Used

- `express.json()`
- CORS configuration
- Error handling middleware

Benefits

- JSON request parsing
- Cross-origin request management
- Centralized error handling
- Improved API reliability

Frontend State Management Review

State management and component synchronization were reviewed.

React Hooks Used

useState

Used for storing and updating component data dynamically.

Example:

```
const [data, setData] = useState([]);
```

useEffect

Used for handling side effects such as API calls and data fetching.

Example:

```
useEffect(() => {  
  fetchData();  
}, []);
```

Benefits

- Dynamic UI updates
- Efficient data synchronization
- Improved user interaction

Refactoring Readiness

The codebase was reviewed to ensure adaptability and maintainability.

Areas Considered

- Endpoint naming consistency
- Variable naming standards
- Route organization
- Component structure
- Code readability

Example Refactoring

Before:

```
app.get("/users", getUsers);
```

After:

```
app.get("/customers", getCustomers);
```

This demonstrates the ability to modify and maintain application logic when requirements change.

Skills Reinforced

During this evaluation and review process, the following concepts were reinforced:

- React Production Builds
- React Router
- useState Hook
- useEffect Hook
- Express.js Middleware
- MongoDB Integration
- API Development
- Environment Variables
- Code Refactoring Practices
- Full Stack Application Structure

Conclusion

Day 6 focused on reviewing and validating the MERN stack application developed throughout the internship. The production build was successfully verified, routing and backend configurations were reviewed, database connectivity was inspected, and frontend state management concepts were reinforced. The codebase was evaluated for maintainability and adaptability through refactoring exercises, improving overall understanding of full-stack application development.